

Bitcoin Has Thin Tails

Modelling Bitcoin Options With Damped Black-Scholes

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Abstract

We examine the distribution of realized Bitcoin daily log-returns and find significantly-thin tails. From there we construct a simple connection back to traditional volatility modelling. And then we discuss how this connection can serve as a foundation to leverage existing derivative quant research to explore cryptocurrency market dynamics. These results also suggest a connection between cryptocurrency exchange structure and trading dynamics.

1 Introduction

Bitcoin has exhibited a high level of realized volatility for over a decade. And this level has varied dramatically over time (Hu et al., 2019). Much of the research in derivative pricing concerns handling assets with significantly non-Normal return distributions with approaches in the style of Heston (1993).

Such work deals with a wide range of parameterizations for volatility, the design of payoff replication strategies that are robust to odd volatility dynamics and techniques for calibrating models that depends on large numbers of unobserved parameters.

Here we have two goals. First, we seek to characterize the nature of Bitcoin's realized volatility distribution. Then, with an understanding of the empirical data, we look to find a simple way to bridge back to traditional financial engineering.

A tremendous range of underlying assets are referenced in traditional derivative markets – it is surely more efficient to find ways to adapt those techniques rather than attempt to reconstruct the system from scratch.

2 Data

We start from public Bitcoin closing price data from mid 2013 through late 2020 from Kumar. For the sample data log return statistics are presented in table 1. Charts of the realized CDF, PDF and dif-

Daily mean	18.3 bp
Daily variance	423bp
Skewness	-0.53
Kurtosis	10.965
Excess kurtosis	-2.01
Lilliefors Statistic	0.1898
Annualized mean	$18.3 \times 365 = 6679\text{bp}$
Annualized variance	$423 \times \sqrt{365} = 6973\text{bp}$

Table 1: Sample Bitcoin log return data with $n=3166$.

ference between a fitted Normal and the realized CDF are given in figures 1, 2 and 3.

This is clearly not Normally distributed data and has significantly thin tails using the test described in Lilliefors (1967). These results are robust to the choice of data source and time range.

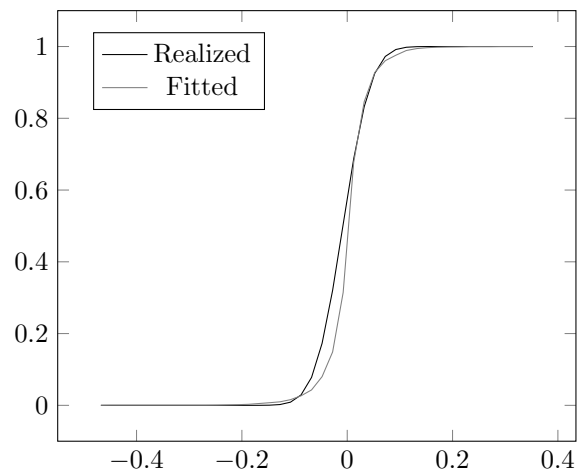


Figure 1: The realized Cumulative Distribution Function compared with the CDF from a Normal distribution with the same mean and variance.

Other research such as Osterrieder and Lorenz (2017) has similarly found that much of the mass in the realized Bitcoin PDF is narrower than Normal. Those authors also note that the mean volatil-

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ity level is far higher than for G10 currency pairs. We note that while Bitcoin realized volatility is higher than the major currencies it is not outside the range of other assets with large option markets where the same quantitative finance techniques are routinely applied. Aggregate US equity market realized volatility was found in [Manda \(2010\)](#) to be around these levels for a prolonged period around the 2008 crisis .

The dataset considered in this paper has a mean annual return volatility of $\sim 70\%$. This level is commonly seen in commodity markets, single-name equities and emerging market currencies experiencing crises. The market for options on Tesla stock is large and currently trades around similar implied volatility levels.

3 Setup

An analysis of historical Bitcoin prices shows they have thin tails. Conventionally financial assets are considered to have fat tails. But let's see if we can leverage some fat-tail work for these thin ones.

In [Lim et al. \(2002\)](#) a model is set up which parameterizes the fatness of the tails. There is a step in that construction of:

$$d \ln(S_t) = (\mu - \frac{\sigma^2}{2})dt + \sigma dz \quad (1)$$

where z is distributed as a generalized Student's t capable of both thin and fat tails. Let's take z' distributed as a standard Normal and $\text{Corr}(z, z') = 1$. Then:

$$d \ln(S_t) = (\mu - \frac{\sigma^2}{2})dt + \sigma(dz - dz') + \sigma dz' \quad (2)$$

And rearrange terms a bit:

$$d \ln(S_t) - \sigma(dz - dz') = (\mu - \frac{\sigma^2}{2})dt + \sigma dz' \quad (3)$$

The right-hand side here is standard Black-Scholes in the style of [Hull \(2006\)](#). The left-hand side is the return plus a correction for non-Normal tails. Recall the processes are perfectly correlated – the left side is a damped log-return.

We now have a simple, albeit limited, mechanism to modify the distribution.

4 Liquidations

One major way in which cryptocurrency trading differs from traditional finance is the prevalence of so-called “liquidations.” This phenomenon is discussed by [Qin et al. \(2021\)](#) and [Soska et al. \(2021\)](#). Cryptocurrency exchange liquidations have the effect of damping day-to-day realized P&L. We know this by construction: when your counterparty is liquidated you face the risk of your profits being subject to a “haircut.” As there is no scenario

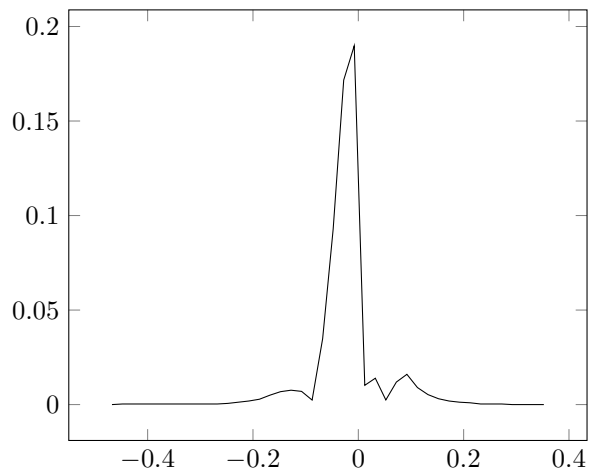


Figure 2: The absolute value of the difference between the realized and fitted CDFs.

where you are paid extra money this has the effect of pulling in the tails.

If we assume the form of the damping is the difference between a Normal and Generalized Student's t what do we get? We get the above formula. We then rearrange – the opposite of what happens above – and find the work is already done.

This is not meant as a glib remark. Much work has been done to manage the attendant risks of non-Normally distributed returns. The sorts of exchange liquidations we see in crypto are not present to the same extent in traditional finance as studied in [FIA \(2018\)](#). Our purpose here is merely to highlight the empirical fact of thin tails and present a plausible mechanism for connecting this to the large body of existing derivative pricing literature.

5 Modified Black Scholes

With this formulation we get the same answer as in [Lim et al. \(2002\)](#). This is because we are assuming that correction as the impact of liquidations. It has the right form and the observed parameters here fit squarely within the values considered in that paper. That entire calibration procedure will carry over.

6 Further Work

The next step is to coordinate with [Alexander et al. \(2021\)](#) to study the impact and levels of liquidations and see if the models can be merged. It seems likely that some combination of short-term mean reversion and what those authors call “smoothing” may also work to explain the dynamics observed here.

It may prove necessary to adopt more complex dynamics in the style of [Hung et al. \(2021\)](#). We note that many, but not all, other fat tail approaches also likely work. This is an area for future study.

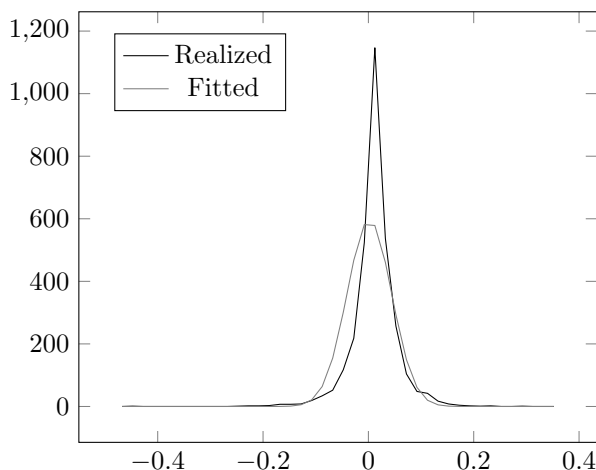


Figure 3: Realized and fitted Probability Distribution Functions. Note both are scaled such that area under the curve is 1.

7 Discussion

This is a simple result. We are merely observing significantly-non-Normal returns for Bitcoin and then reverse engineering a plausible-and-already-solved model for the impact of haircuts in the data.

There is still much work to do in comparing empirical exchange liquidations against a range of plausible return distributions. But this gives a place to start by connecting traditional volatility modelling and empirical evidence of idiosyncratic cryptocurrency trading dynamics.

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